

TrES-2b

R-2356

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_101111 · created 2026-08-19T22:48:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455511.651389 +/- 4.4e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.014
Transit depth (Rp/Rs)^2	1.25 +/- 0.32 [%]
Orbital Inclination (inc)	85.71 +/- 1.45
Airmass coefficient 1 (a1)	1.314 +/- 0.011
Airmass coefficient 2 (a2)	-0.1131 +/- 0.0035
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	None
Optimal Aperture	3.47
Optimal Annulus	7.22
Transit Duration (day)	0.091 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.112 ± 0.014 vs 0.1254 (deviation 0.77σ)
Duration fit vs published	0.091 d vs 0.0746 d (deviation 0.88σ)
Mid-time O–C	0.1 min
Depth SNR	6.5
Residual scatter	0.8 %

Light curve

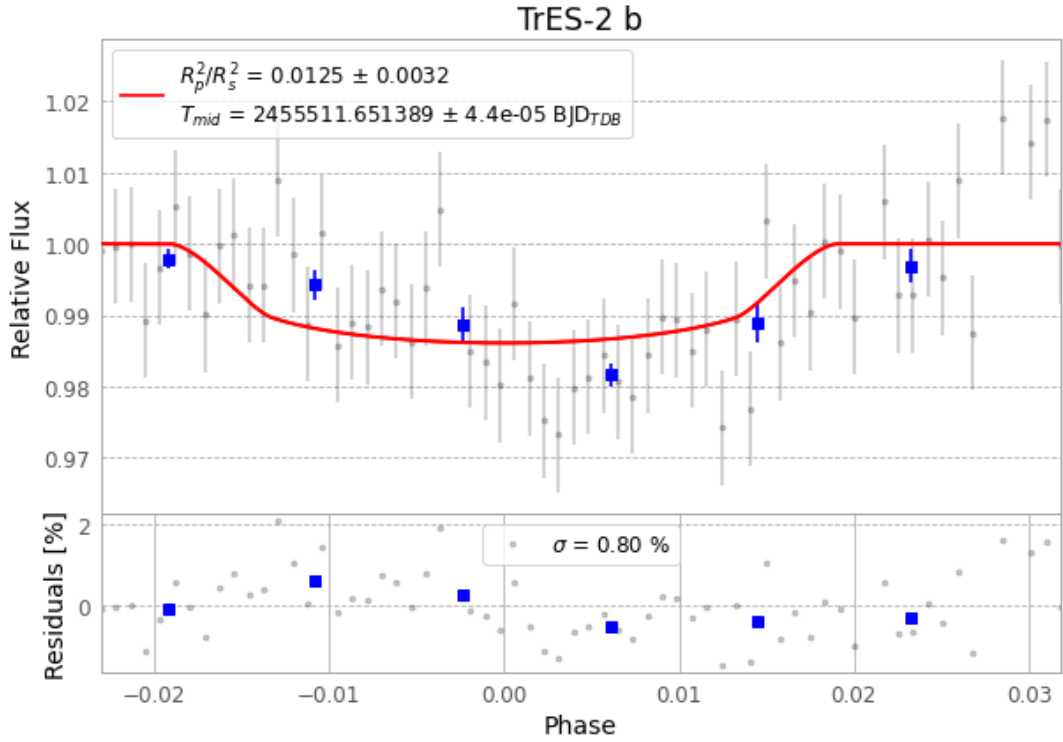


Figure 1 — FinalLightCurve_TrES-2 b_2010-11-10.png (SHA-256 7de30b64d7d39db2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96a415ffcc0ebb62...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

62 FITS frames (41 MB), TRES-2101111021513.FITS → TRES-2101111053014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-101111.log (23272ba8f10c...), FinalLightCurve_TrES-2 b_2010-11-10.png (7de30b64d7d3...), FinalLightCurve_TrES-2 b_2010-11-10.csv (edb21ef438d8...)

Compute resources

Wall time 126.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 22:48Z.